

single point of failure because of how a distributed system operates. Even if one node fails, the remaining nodes continue to operate, ensuring no disruption.

- **Immutable:** All the transactions are immutable in a blockchain technology based system. Encryption is done for every transaction covering the time, date, the participants and the hash to the previous block.
- **Chronology:** Each block acts like a repository that stores information pertaining to a transaction and links to the previous block in the same transaction. These connected blocks form a chronological chain providing a trail of the underlying transaction.
- **Consensus based:** A transaction on blockchain is executed only if all the parties on the network unanimously approve it. Also, consensus based rules can be altered to suit various circumstances.
- **Digital signature:** Blockchain enables the exchange of transactional value using unique digital signatures that rely on public keys (decryption code known to everyone on the network). Private keys are codes known only to the owner to create proof of ownership. This is very critical in avoiding fraud in record management.
- **Consistent:** Blockchain data is complete, consistent, timely, accurate, and widely available.
- **Persistence:** Blockchain will not create/persist invalid transactions as determined by consensus. It is nearly impossible to delete or roll back transactions once they are included in the blockchain. Cryptographically, the blocks created are sealed in the chain. It is impossible to delete, edit or copy already created blocks and put them on the network. This leads to the creation of digital assets and ensures a high level of robustness and trust.
- **Anonymity:** Each user can interact with the blockchain with a generated address, which does not reveal the real identity of the user.

Security features of blockchain

Blockchain helps in recording transactions of any digital assets exchanged between two unknown parties. Security aspects supported by blockchain are critical in ensuring transparency, confidentiality and protection against fraud. The following are the high-level security features of the blockchain.

Ledger: The ledger records every transaction in the blockchain. The ledger is a chain of blocks and information in the block is immutable. The distribution of the ledger is done to all the nodes.

Chain of blocks: Blockchain is a chain of blocks. Each block has the hash value of the previous block and this forms a chain. Correction to data in a block (say, n) will change the hash value and will not validate with the hash stored in the next block (n+1). This will be a chain reaction and affect the overall chain. Therefore, this characteristic increases the protection of sensitive data or information.

Confidentiality: Blockchain provides confidentiality by enabling users of a ledger to see authorised transactions only.

Transparency: Blockchain allows the transactions and the ledger state to be maintained and be managed transparently by sharing the ledger to all nodes and using consensus algorithms to reach consensus among all nodes. Consensus algorithms also ensure the ordering and execution of the transactions.

Cryptology: This enables secure transactions and makes blockchain immutable using hash based algorithms, which produce a fixed hash, based on the content of the block.

Smart contracts: A smart contract is a computer code running on top of a blockchain containing a set of rules, on the basis of which the parties agree to interact with each other. If the pre-defined rules are fulfilled, then the agreement is automatically executed. No contract will execute without the network consensus.

Security reference architecture in the blockchain

Blockchain security ensures that the right people, internal or external, get access to the appropriate data and information at the right time and place, within the right channel. Security prevents and safeguards against malicious attacks; it protects enterprise data assets by securing and encrypting data while it is in motion or at rest. It also enables organisations to separate roles and responsibilities, protecting sensitive data without compromising privileged user access.

The security at various layers for an enterprise application is classified as:

- User access layer security
- Presentation layer security
- Identity and access management layer security
- Application/data layer security
- Network layer security
- Infrastructure layer security

Figure 1 gives a layered view of the security approach with blockchain.

User access layer: Various stakeholders, both internal and external, are part of this layer. They are the primary users of the systems. Stakeholders use channels to interact with the enterprise. They engage with various departments or business units of the enterprise over multiple channels, both physical and digital.

Presentation layer: The front-end application security should ensure the following.

- **Authentication:** Authentication is the assertion by a subscriber to prove his/her identity. The authenticators and factors are:
 - Something that one has to show (access card)
 - Something one knows (password) and
 - Something one is (biometric).

Authentication mechanisms should be commensurate with the strength of the identity model, the level of access and the sensitivity of the transaction. They must implement a combination of multiple authenticators (*aka* multi-factor authentication).